

## Title of Micro Project: - Indian Defence Management System

### TABLE CREATION & DATA INSERTION

- **RANK TABLE**

CREATE TABLE Rank

```
(  
rank_id  NUMBER PRIMARY KEY,  
rank_name VARCHAR2(50) NOT NULL  
);
```

INSERT INTO Rank VALUES (1, 'Private');

INSERT INTO Rank VALUES (2, 'Sergeant');

INSERT INTO Rank VALUES (3, 'Captain');

INSERT INTO Rank VALUES (4, 'Major');

SELECT \* FROM Rank;

- **UNIT TABLE**

CREATE TABLE Unit

```
(  
unit_id  NUMBER PRIMARY KEY,  
unit_name VARCHAR2(50) NOT NULL,  
location VARCHAR2(100) NOT NULL  
);
```

INSERT INTO Unit VALUES (101, 'Northern Command', 'Jammu & Kashmir');

INSERT INTO Unit VALUES (102, 'Eastern Command', 'Assam');

INSERT INTO Unit VALUES (103, 'Naval Fleet Alpha', 'Mumbai');

SELECT \* FROM Unit;

- **SOLDIER TABLE**

```
CREATE TABLE Soldier
(
soldier_id NUMBER PRIMARY KEY,
name    VARCHAR2(50) NOT NULL,
age     NUMBER,
rank_id  NUMBER,
unit_id  NUMBER,
CONSTRAINT fk_soldier_rank FOREIGN KEY (rank_id) REFERENCES Rank(rank_id),
CONSTRAINT fk_soldier_unit FOREIGN KEY (unit_id) REFERENCES Unit(unit_id)
);

INSERT INTO Soldier VALUES (1001, 'Rahul Singh', 25, 1, 101);
INSERT INTO Soldier VALUES (1002, 'Arjun Mehta', 28, 2, 101);
INSERT INTO Soldier VALUES (1003, 'Vikram Rao', 32, 3, 102);
INSERT INTO Soldier VALUES (1004, 'Karan Kapoor', 40, 4, 103);

SELECT * FROM Soldier;
```

- **OPERATION TABLE**

```
CREATE TABLE Operation
(
operation_id  NUMBER PRIMARY KEY,
operation_name VARCHAR2(100) NOT NULL,
start_date   DATE,
end_date     DATE
);

INSERT INTO Operation VALUES (201, 'Operation Vijay', TO_DATE('1999-05-01', 'YYYY-MM-DD'),
TO_DATE('1999-07-26', 'YYYY-MM-DD'));

INSERT INTO Operation VALUES (202, 'Operation Trident', TO_DATE('1971-12-04', 'YYYY-MM-DD'),
TO_DATE('1971-12-06', 'YYYY-MM-DD'));

INSERT INTO Operation VALUES (203, 'Operation Parakram', TO_DATE('2001-12-15', 'YYYY-MM-DD'),
TO_DATE('2002-10-16', 'YYYY-MM-DD'));
```

```
SELECT * FROM Operation;
```

- **PARTICIPATION TABLE**

```
CREATE TABLE Participation
```

```
(
```

```
soldier_id  NUMBER,
```

```
operation_id NUMBER,
```

```
role        VARCHAR2(50),
```

```
CONSTRAINT pk_participation PRIMARY KEY (soldier_id, operation_id),
```

```
CONSTRAINT fk_participation_soldier FOREIGN KEY (soldier_id) REFERENCES Soldier(soldier_id),
```

```
CONSTRAINT fk_participation_operation FOREIGN KEY (operation_id) REFERENCES Operation(operation_id)
```

```
);
```

```
INSERT INTO Participation VALUES (1001, 201, 'Infantry');
```

```
INSERT INTO Participation VALUES (1002, 201, 'Medic');
```

```
INSERT INTO Participation VALUES (1003, 202, 'Commander');
```

```
INSERT INTO Participation VALUES (1004, 203, 'Naval Officer');
```

```
SELECT * FROM Participation;
```

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## ➤ Queries

- COMMIT  
COMMIT;
  - SAVEPOINT  
SAVEPOINT before\_update;
  - UPDATE  
UPDATE Soldier SET age = 26 WHERE soldier\_id = 1001;
  - ROLLBACK TO SAVEPOINT (undo update)  
ROLLBACK TO before\_update;
  - DELETE  
DELETE FROM Soldier WHERE soldier\_id = 1002;
  - ROLLBACK (undo delete)  
ROLLBACK;
- 

- **DCL**

GRANT SELECT, INSERT ON Soldier TO trainee\_user;  
REVOKE INSERT ON Soldier FROM trainee\_user;

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- **STORED PROCEDURE**

```
CREATE OR REPLACE PROCEDURE add_soldier
(
  p_id NUMBER, p_name VARCHAR2, p_age NUMBER, p_rank NUMBER, p_unit NUMBER
) AS
BEGIN
  INSERT INTO Soldier VALUES (p_id, p_name, p_age, p_rank, p_unit);
  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Soldier added successfully: ' || p_name);
END;
/
```

---

- **FUNCTION**

```
CREATE OR REPLACE FUNCTION get_soldier_count(p_unit NUMBER) RETURN NUMBER IS
v_count NUMBER;
BEGIN
  SELECT COUNT(*) INTO v_count FROM Soldier WHERE unit_id = p_unit;
  RETURN v_count;
END;
/
```

---

- **TRIGGER**

```
CREATE OR REPLACE TRIGGER trg_check_age
BEFORE INSERT OR UPDATE ON Soldier
FOR EACH ROW
BEGIN
  IF :NEW.age < 18 THEN
    RAISE_APPLICATION_ERROR(-20001, 'Soldier age must be at least 18.');
```

```
  END IF;
END;
/
```

---

## ➤ JOIN QUERIES

- **INNER JOIN: Soldier with Rank and Unit**

```
SELECT s.soldier_id, s.name, r.rank_name, u.unit_name
FROM Soldier s
JOIN Rank r ON s.rank_id = r.rank_id
JOIN Unit u ON s.unit_id = u.unit_id;
```

- **LEFT JOIN: Soldier count per unit**

```
SELECT u.unit_name, COUNT(s.soldier_id) AS soldier_count
FROM Unit u
LEFT JOIN Soldier s ON u.unit_id = s.unit_id
GROUP BY u.unit_name;
```

- **JOIN with filter: Soldiers in Operation Vijay**

```
SELECT s.name, p.role
FROM Soldier s
JOIN Participation p ON s.soldier_id = p.soldier_id
JOIN Operation o ON p.operation_id = o.operation_id
WHERE o.operation_name = 'Operation Vijay';
```

---

- **DROP TABLE**

```
DROP TABLE Participation CASCADE CONSTRAINTS;
```